

EXAM REVIEWER: CLOUD COMPUTING

COMPREHENSIVE ALL-IDENTIFICATION MATRIX (CHAPTERS 1–7)

Identification Practice Format: Use this document to master definitions. Test yourself by covering the rightmost column ("Concept / Answer Key") and identifying the technical term based on the definition provided.

CHAPTER 1: WHAT IS CLOUD COMPUTING?

DEFINITION / CLUE	CONCEPT / ANSWER KEY
A facility used by organizations to house and manage data, including hardware components like compute, storage, and networks, and software components like applications and operating systems.	Data Center
A person or organization requesting cloud resources and services (alternately referred to as user, client, customer, or consumer).	Cloud User
An organization that provides a cloud platform, infrastructure, applications, and services.	Cloud Service Provider (CSP)
A core cloud architecture where multiple customers share the available physical resources of a cloud provider without the knowledge or visibility of one another.	Multitenancy
Applications or services that are specifically designed, built, and optimized to run natively within a cloud computing environment.	Cloud-native
A situation where a customer relies on proprietary software or technology that restricts them from shifting to alternative solutions without significant cost or effort.	Vendor lock-in
The economic relationship where an increase in production volume leads to a decrease in per-unit cost by spreading out fixed expenses over more units.	Economies of scale
The capacity of an enterprise to rapidly alter its operations and adapt to sudden fluctuations in the market or industry.	Organizational Agility
Costs associated with purchasing, improving, or maintaining fixed physical assets (e.g., buying land, servers, or permanent software licenses).	Capital Expenses (CAPEX)
Costs tied to the day-to-day running of a business or infrastructure (e.g., monthly Internet connection costs, software licensing subscriptions).	Operational Expenses (OPEX)

DEFINITION / CLUE	CONCEPT / ANSWER KEY
A scaling mechanism that involves adding resources like memory or processing power to the exact same computing node.	Vertical scaling
A scaling mechanism that involves adding more separate nodes or machines to a distributed system architecture.	Horizontal scaling
The complete financial estimate of an object or service throughout its entire lifecycle, from acquisition to disposal, including hidden direct and indirect costs.	Total Cost of Ownership (TCO)
An economic or business term representing obstacles or hurdles that make it highly difficult for a new company to enter a given market segment.	Barrier to entry
An application architectural pattern where the system is built and deployed as a single, indivisible, tightly coupled unit.	Monolithic application
An application pattern where the system is broken down into a suite of small, independent, and loosely-coupled services communicating via APIs.	Microservices-based application
A model that abstracts server management, allowing developers to deploy code independently without the administrative burden of provisioning or managing persistent servers.	Serverless
The on-demand delivery of servers, storage, databases, applications, and IT resources over the Internet, where details of system implementation are abstracted from the user.	Cloud Computing
The US agency that produced Special Publication 800-145, establishing the widely accepted standard framework containing 5 characteristics, 4 deployment models, and 3 service models.	NIST
The NIST characteristic where a cloud consumer can provision computing assets automatically as needed without requiring human interaction with the CSP.	On-demand Self Service

CHAPTER 2: STORAGE AND DATABASES

DEFINITION / CLUE	CONCEPT / ANSWER KEY
Data that provides descriptive context, properties, or attributes about other data (commonly simplified as "data about data", such as owner, size, or creation date).	Metadata
The deep structural integration of digital technology into all functional areas of a business, altering how it operates and delivers value.	Digital Transformation

DEFINITION / CLUE	CONCEPT / ANSWER KEY
The fundamental unit of measurement used to quantify computing storage capacity.	Bytes
The systematic practice of identifying old or inactive data that is no longer in active use and storing it securely long-term for regulatory or compliance reasons.	Data archiving
The metrics-driven property of a system to remain continuously available and operational without service disruption (e.g., 99.99% availability allows only 52m 35s of downtime per year).	High Availability (HA)
The formal blueprint, layout, configuration, or structural organization of data fields and tables within a relational database system.	Schema
A database scaling technique that involves horizontally partitioning a massive database table into smaller, separate, and easily managed portions.	Sharding
A traditional, high-performance storage framework where files are split into raw, equal-sized chunks without metadata, optimal for transactional databases.	Block Storage
A cloud storage architecture where files and data are kept in a flat address space along with custom metadata and a unique identifier.	Object Storage

CHAPTER 3: VIRTUALIZATION, VIRTUAL MACHINES, AND MORE

DEFINITION / CLUE	CONCEPT / ANSWER KEY
The low-level legacy firmware system that allows a computer or server motherboard to initialize hardware and boot up when powered on.	BIOS (Basic Input/Output System)
The modern, feature-rich firmware interface that replaced BIOS in 2007, offering substantially faster boot times and enhanced hardware support.	UEFI
Any application, program, service, or computing task that actively runs on a system and consumes physical underlying resources.	Workload
A single physical processor unit that contains two or more completely independent processing units (cores) to facilitate simultaneous multitasking.	Multicore processor
An Intel-proprietary hardware optimization technology that tricks the operating system into recognizing two separate logical cores for each physical core processor (pCPU).	Hyperthreading
The virtualization strategy of assigning and allocating more virtual infrastructure resources (vCPU, vRAM) to guests than what is physically present on the host.	Overcommitment / Oversubscribing

DEFINITION / CLUE	CONCEPT / ANSWER KEY
The foundational enabling technology of cloud computing that abstracts physical hardware, allowing a single host to run multiple virtual machines concurrently.	Virtualization
An isolated, logical software-defined computing environment that bundles its own guest operating system and applications, functioning exactly like a standalone physical server.	Virtual Machine (VM)

CHAPTER 4: NETWORK INFRASTRUCTURE AND VIRTUAL NETWORKS

DEFINITION / CLUE	CONCEPT / ANSWER KEY
A completely private network infrastructure that remains entirely under the administrative control of a single organization, restricted exclusively to internal staff.	Intranet
A controlled extension of an internal private network that grants safe, restricted access to verified external entities like suppliers, vendors, or business partners.	Extranet
The global, publicly accessible, interconnected network of computers that operates in an unmanaged, unregulated fashion.	Internet
A highly scalable network architecture where distributed endpoint systems (clients) explicitly request files and processing tasks from a centralized host.	Client-server architecture
A small-scale network architecture where individual nodes interact as equals, alternating roles between client and server based on active sharing needs.	Peer-to-peer architecture
A network that spans a restricted geographic territory like an office or single corporate building, governed by the IEEE 802.3 Ethernet standard.	Local Area Network (LAN)
A long-distance network architecture that interconnects disparate local networks across massive geographic regions, frequently utilizing leased lines.	Wide Area Network (WAN)
A localized network that bridges high-speed data connections across a unified academic campus or metropolitan city landscape.	Metropolitan Area Network (MAN)
A local area network that implements wireless connectivity parameters based on established IEEE 802.11 standards.	Wireless LAN (WLAN)
The specific networking device or endpoint that acts as either the primary source or the ultimate destination of an active network communication.	End Device

CHAPTER 5: CLOUD SECURITY

DEFINITION / CLUE	CONCEPT / ANSWER KEY
The cryptographic process of running mathematical algorithms and keys to turn original clear text into an unreadable, encoded layout.	Encryption
Original, raw, and unencrypted data elements before any cryptographic algorithms have been applied to encode them.	Plaintext
The scrambled, unreadable, and secure text output that emerges immediately after encryption processes are completed.	Ciphertext
A dynamic firewall filtering mechanism that continuously tracks connection states and permits return packets only if they match a valid internal request.	Stateful Packet Inspection (SPI)
The definitive security architectural framework delineating that the cloud provider secures the underlying physical infrastructure, while the client secures data and access configurations.	Shared Responsibility Model
The operational process of running scans to discover flaws or system code structural weaknesses across 3 core phases: intelligence gathering, assessment, and validation.	Vulnerability scanning
An advanced, active security evaluation mechanism that models real-world exploits by actively attacking target environments across 7 detailed deployment phases.	Penetration testing

CHAPTER 6: BUSINESS CONTINUITY AND DISASTER RECOVERY

DEFINITION / CLUE	CONCEPT / ANSWER KEY
Any unplanned or abrupt event caused by human actions, technical errors, or natural phenomena that induces partial or absolute operational disruption to a business.	Disaster
The precise operational process of immediately redirecting active production workloads over to a secondary recovery site when a primary system goes down.	Failover
The operational procedure of restoring production workloads back to their primary server infrastructure or data site once formal repairs and testing are certified.	Failback
The comprehensive macro-level operational methodology and policy structure designed to ensure core corporate functions remain active during a major disruptive crisis.	Business Continuity (BC)

DEFINITION / CLUE	CONCEPT / ANSWER KEY
A technical subset of business continuity that focuses explicitly on the strategies and mechanics required to restore systems and operations back to normal following a disaster.	Disaster Recovery (DR)

CHAPTER 7: TESTING, AUTOMATION, AND DEVOPS

DEFINITION / CLUE	CONCEPT / ANSWER KEY
The foundational dataset or benchmark values gathered over extended sampling intervals that map out what constitutes normal system performance and activity.	Baseline
The programmatic methodology of managing and provisioning foundational infrastructure tracks via declarative, automated configuration templates.	Infrastructure as Code (IaC)
A deployment strategy where new software updates are applied directly onto existing nodes, causing service outages because old instances must stop before updates start.	In-place deployment
A sequential deployment model where the software update is applied instance-by-instance across the infrastructure to prevent total environment outages.	Rolling deployment
An update strategy utilizing two identical hosting environments: one active environment handles production traffic while the other tests the new software before traffic is flipped.	Blue/Green deployment
A phased deployment technique where software updates are initially introduced to a tiny percentage of live users before rolling out linearly to the rest of the infrastructure.	Canary deployment